

1(a)(i)	<i>answers must be comparative</i> <i>one from:</i> <i>idea that</i> the sections are orientated differently / cut in different planes / cut at different angles / AW ; A is a cross section / AW, and B is a longitudinal section / AW mitochondria show a variety of, sizes / shapes ; mitochondria, are flexible / change shape ; A and B are of, different ages / stages of development ;	1
1(a)(ii)	<i>two from:</i> replace, old / worn out / damaged, mitochondria ; require more mitochondria as cells, enlarge / grow ; more mitochondria are needed, for new cells / when cell divides ; A so numbers remain the same in the new cells / after cell division A idea that mitochondria are shared out (between daughter cells) (mitochondria) provide (most), energy / ATP ; I cell needs (much) energy without stating mitochondria provide energy R energy production I uses of energy	2
1(b)	<i>two from:</i> nuclear envelope shown as two membranes with closed ends and a gap ; label to nuclear pore ;	2

Question	Answer	Marks
1(c)	<i>four from:</i> 1 contains / AW, chromosomes / chromatin / DNA / genes / genetic material / inherited material / genetic information ; 2 (coded) information for synthesis of, polypeptides / proteins ; A controls protein synthesis 3 ref. to sequence(s) of bases (in DNA) ; 4 transcription (of genes) / production of mRNA ; 5 contains nucleolus ; 6 manufacture of ribosomal sub-units ; A makes ribosomes 7 <i>idea that</i> DNA is protected from degradation / enzymes / AW ; 8 AVP ; e.g. responds to signals that control, gene expression / AW post-transcription modification / modifies mRNA / any example repairs DNA I controls cell's activities I sends mRNA to cytoplasm / mRNA travels through nuclear pores	4

Question	Answer	Marks
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